Post Replied to

by Bashair Alhosani - Monday, 17 August 2026, 9:06 AM
Number of replies: 2

Aviation has been an enthusiastic adopter of Industry 4.0, deploying cyber-physical systems, the Internet of Things (IoT) and big-data analytics across predictive maintenance, dynamic pricing and flight operations (Lasi et al., 2014). However, Metcalf (2024) argues that Industry 5.0 emerged partly as a reaction to the excesses of this techno-economic paradigm, proposing a socio-technical future in which human-centricity, resilience and sustainability determine how technology is deployed (Breque, De Nul and Petridis, 2021; Xu et al., 2021). The aviation sector's continued dependence on ageing information systems illustrates why this shift matters.

A striking example is Southwest Airlines' operational collapse of December 2022. When Winter Storm Elliott disrupted flights across the United States, most carriers recovered within days; Southwest did not, because its legacy crew-scheduling system (SkySolver) could not re-match crews to aircraft at scale, forcing staff into slow manual rescheduling (US Department of Transportation, 2023). The airline cancelled approximately 16,700 flights over ten days and stranded more than two million passengers (BBC News, 2023). The direct economic cost exceeded US$1.1 billion in refunds, reimbursements and lost ticket sales (CBS News, 2023), and in December 2023 the US Department of Transportation (2023) imposed a record US$140 million civil penalty for consumer-protection violations. The reputational damage was intensified by revelations that employee unions had repeatedly warned management about the outdated scheduling technology (CNN, 2023).

This incident demonstrates that Industry 4.0 adoption in aviation has been uneven: customer-facing digitalisation outpaced investment in core operational systems, creating fragility rather than the resilience that Industry 5.0 explicitly demands (Xu et al., 2021). Indeed, Metcalf (2024) observes that optimising for efficiency can erode resilience, because a maximally efficient system retains no spare capacity to absorb disruption — precisely the position in which Southwest's lean, point-to-point operation found itself when its digital infrastructure failed. It also highlights the human dimension of systems failure — overwhelmed call centres and crews unable to reach schedulers transformed a technical fault into a humanitarian and reputational crisis. For aviation, the lesson is that technical debt in mission-critical information systems constitutes a strategic risk, and that resilient, human-centric system design should be a board-level priority rather than merely an IT concern.

I would welcome peers' views on whether regulatory penalties alone are sufficient to drive legacy-system modernisation, or whether genuine resilience requires the cultural and organisational change envisaged by Industry 5.0.

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My Response

Dear Bashair,

Thank you for your post. You make some interesting points and ask for your peers' points of view regarding regulatory penalties versus cultural and organisational change.

My view is that regulatory penalties are often disregarded by companies as, in order to incur the penalty, a catastrophic event would need to take place and a penalty such as a fine would be the least of their worries.

Hardy, T. (2020) argues that Orthodox Deterrence theory is often not effective as companies do not expect to be caught out and that members of that organisation are often acting in their own interests, rather than the company's, so deterrents need to work on a personal level and not just a corporate level.

Your post highlights that what was an unpredictable event, became a catastrophic event for Southwest Airlines because they did not have resilience built into their disaster recovery systems. Either they were unconcerned about regulatory penalties, or they simply under-invested in emergency planning and disaster recovery procedures.

Other airlines recovered swiftly, proving that resilience was possible. An event such as this would encourage Southwest Airlines to engage with other airlines in order to implement the cultural and organisational changes that would enable them to build such resilience.

My theory is that examples such as this encourage industry operators to benchmark themselves against other organisations and, by sharing best practices, they can quickly develop resilience in the same way that the technologies have been developed though collaboration. Regulatory penalties on the other hand, are things that organisations hope they'll never need and it's easier to ignore these than it is to watch competitors increase productivity and cope with minor issues without attempting to replicate that robustness.

References

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